Research Plan — Revised Draft v5

Title (Primary):

> *Modeling Factors Influencing Student Adoption of AI-Based Adaptive Learning Platforms: A Hybrid SEM–ANN Approach Among Higher Education Students in Bangladesh*

Short Title (if journal requires): Student Adoption of AI-Adaptive Learning Platforms: A Hybrid PLS-SEM–ANN Study

Target: Q1 Journal Publication + BSc Final Defense **Preferred Journals:** *Computers & Education: Artificial Intelligence* (Elsevier) | *Education and Information Technologies* (Springer) **Method:** PLS-SEM (SmartPLS 4) + ANN (Python / SPSS Neural Networks)

> **Revision note (v5):** *Built on v4. New fixes in this version: (1) SmartPLS* **software citation** *added — Ringle, Wende & Becker (2024) required by journals alongside the Hair et al. (2022) methodological reference; (2) ANN activation function corrected —* **sigmoid replaced with ReLU** *for hidden layers and* **linear** *for the output layer; sigmoid is outdated and causes vanishing gradients; (3)* **PLSpredict** *added to structural model assessment — now a Q1 standard for out-of-sample predictive validity; (4)* **PLS-SEM vs. CB-SEM justification** *added to the methodology section; (5)* **perception vs. experience gap** *addressed in survey introduction and limitations; (6)* **ANN predictive accuracy metrics** *(R^2 and RMSE) added to reporting plan; (7)* **non-response bias check** *(Armstrong & Overton, 1977) added to Phase 5; (8)* **IRB terminology** *replaced with "departmental research ethics committee" appropriate to Bangladeshi university context; (9)* **CB-SEM robustness check** *flagged as an optional step if requested by reviewers; (10) note added about the forthcoming Hair et al. 4th edition (2026); (11) Singh & Zhang (2022) verification note added to references. See Section 15 for the full change log across all versions.*

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1. Background

AI-based adaptive learning platforms are fundamentally changing higher education by moving away from static, one-size-fits-all content delivery toward genuinely personalized learning experiences. Unlike traditional e-learning systems (e.g., Moodle, Google Classroom), these platforms use machine learning algorithms to continuously build an individualized learner model — tracking quiz performance, error patterns, and learning pace — and autonomously adjust content sequencing, difficulty, and feedback in real time.

The global EdTech market is growing rapidly. According to HolonIQ's 2023 AI in Education Survey, 25% of educational organizations reported successful AI deployment in 2022, up from 14% in 2019, with the fastest growth in analytics-driven adaptive learning. Post-2023, the generative AI boom alongside sustained post-pandemic digital learning habits has dramatically accelerated institutional interest in AI-adaptive systems. Universities worldwide are investing in platforms like ALEKS and Knewton Alta to improve learning outcomes. Yet despite institutional investment, **actual student adoption and sustained engagement remain low.** Many students disengage after initial exposure due to trust concerns, perceived complexity, privacy worries, or a sense that the system is a "black box" they do not understand (Bozkurt et al., 2024; Gligorea et al., 2023).

In Bangladesh, higher education institutions — including public universities, private universities, and colleges under the National University — are gradually integrating digital and AI-assisted learning tools. According to the Bangladesh University Grants Commission (UGC) annual reports, the country has over 4 million university-enrolled students, the vast majority of whom interact with digital learning tools primarily through smartphones. Local platforms such as 10 Minute School and Bohubrihi represent conventional e-learning environments: primarily video-based and instructor-led, without the autonomous AI-driven content adaptation that defines the platforms studied here. The absence of such AI-adaptive systems in the local EdTech landscape makes **student-side adoption of full AI-adaptive course environments poorly studied in the Bangladeshi context**, which is an important gap in both regional and global literature.

This study focuses specifically on **student adoption** (not teacher or institutional adoption) because the ultimate success of any EdTech investment depends on whether students actually engage with the system regularly and willingly.

> **Action item before defense:** *Confirm exact UGC enrollment statistics from the most recent annual report and add one statistic on AI tool usage rates among Bangladeshi university students if available from a local survey or BANBEIS data.*

2. Problem Statement & Introduction

Although AI-adaptive platforms have demonstrated measurable benefits — improved retention, higher engagement, and better learning outcomes in controlled studies (Bozkurt et al., 2024; Singh & Zhang, 2022) — **real-world adoption by students is inconsistent and often poor.** Most existing research on technology adoption in education either:

- Studies **general e-learning** or LMS tools (Moodle, Blackboard) using standard TAM/UTAUT without AI-specific constructs,
- Examines **simple AI tools** like chatbots or generative AI assistants (ChatGPT), which differ fundamentally from adaptive course environments, or
- Focuses on **teacher/institutional adoption** rather than the student perspective.

Very few studies examine why students choose or refuse to adopt **course-like AI-adaptive platforms** — systems that do not just provide information but actively make decisions about what a student should learn next. This distinction matters because the user dynamics are qualitatively different: the system monitors the student, evaluates them continuously, and controls their learning path. This introduces AI-specific psychological factors — such as anxiety about being evaluated, concern about opaque algorithmic decisions, and sensitivity to whether the system truly personalizes — that generic e-learning adoption models do not capture.

Furthermore, most existing adoption studies use only SEM (assuming linear relationships between constructs), which may miss the complex, nonlinear dynamics of human adoption behavior. A hybrid SEM–ANN approach addresses this limitation by combining the confirmatory hypothesis-testing power of PLS-SEM with the nonlinear predictive importance ranking of artificial neural networks.

Research Objectives:

1. Identify and test the significant factors influencing students' behavioral intention to adopt AI-based adaptive learning platforms in Bangladesh.
2. Apply a hybrid PLS-SEM + ANN methodology to both test theoretical hypotheses and rank predictors by nonlinear importance.
3. Provide actionable recommendations for platform developers, educators, and policymakers.

Research Questions:

1. Which factors significantly predict student behavioral intention to adopt AI-adaptive learning platforms among higher education students in Bangladesh?
2. Which construct has the highest predictive importance (via ANN normalized importance / sensitivity analysis)?
3. Do AI-specific factors (perceived personalization, AI transparency, AI anxiety, feedback quality) explain adoption intent above and beyond standard UTAUT2 constructs alone?

> **Context paragraph for Bangladesh (add before submission):** *Bangladesh's higher education sector is undergoing rapid digital transformation, supported by the government's "Digital Bangladesh" and subsequent "Smart Bangladesh" initiatives. Despite significant investment in EdTech infrastructure, adoption of AI-powered learning tools among students remains nascent and understudied. This study addresses that gap by providing the first empirically validated adoption model specific to AI-adaptive learning platforms in a Bangladeshi higher education context.*

3. Operational Definition

> *Use this definition consistently throughout your thesis and paper. Never substitute "AI tool," "AI chatbot," or "e-learning platform" — this specificity is the core of your novelty argument.*

AI-based adaptive learning platforms, as operationalized in this study, are *structured digital learning environments that employ machine learning algorithms to continuously build and update an individualized learner model — capturing a student's knowledge state, error patterns, and learning pace — and use this model to dynamically adjust content sequencing, difficulty levels, and feedback in real time.*

Unlike generative AI tools such as standalone chatbots (e.g., ChatGPT), these platforms function as **course-like environments** in which students progress through lessons, quizzes, and adaptive activities within a defined curriculum. The defining characteristic is the system's **autonomous decision-making** about what a student should learn next, based on prior performance data rather than direct student choice. This applies whether the learner uses the platform within a formal university course or as a self-directed learner outside formal academic settings.

Examples: ALEKS (McGraw-Hill), Knewton Alta (Wiley), Duolingo's adaptive engine.

4. Platform Examples

Include a brief description of 2–3 of these in your survey introduction so participants have a clear mental model of what they are being asked about.

Platform	Developer	How AI Adapts	Context
ALEKS	McGraw-Hill	Builds a knowledge map per student; continuously assesses and adjusts content	University STEM, most widely used
Knewton Alta	Wiley	Real-time difficulty adjustment and targeted content recommendations	University STEM courses
Duolingo	Duolingo Inc.	Spaced repetition + ML; adjusts lesson difficulty based on error patterns	Language learning (university & self-study)

> **What distinguishes these from regular e-learning:** *A regular LMS (Moodle, Google Classroom) delivers the same content to all students in a fixed sequence. AI-adaptive platforms give each student a different content sequence, different difficulty levels, and personalized feedback — automatically, without instructor intervention. This applies across formal coursework and self-directed use.*

5. Gap Analysis

What Exists in the Literature

- Hundreds of TAM/UTAUT studies on general e-learning and LMS platforms.
- A growing body of research on ChatGPT and generative AI tool adoption (e.g., Algerafi et al., 2023; Boubker, 2024).
- A small but emerging set of hybrid SEM–ANN papers in EdTech (mostly post-2022).
- Studies on AI adoption in education generally — intelligent tutors, recommendation systems, adaptive platforms — but rarely integrating AI-specific psychological constructs.

What Is Missing (Your Research Gaps)

Gap	What Prior Work Does	What This Study Does
Platform specificity	Studies "AI tools" broadly (chatbots, generative AI)	Focuses specifically on course-like AI-adaptive platforms
Perceived Personalization	Not included in any standard adoption model	Included as a novel, adaptivity-specific construct
AI Transparency	Rarely studied in education adoption models	Included as an AI-specific construct grounded in XAI literature (Shin & Park, 2019; Shin, 2021)
AI Anxiety in adaptive learning	Studied occasionally, only with linear SEM	Included with ANN to reveal possible nonlinear/threshold effects; operationalized specifically as surveillance/profiling anxiety in adaptive learning contexts
Feedback Quality	Absent from AI adoption models	Included as an adaptivity-specific construct grounded in adaptive learning systems research
SEM–ANN in adaptive learning	SEM–ANN used for general e-learning, LMS, cloud services	First SEM–ANN study specifically for AI-adaptive platforms
Bangladesh context	Developing/South-Asian contexts underrepresented in AI-adoption literature	Focuses on Bangladeshi university students

> **Selling point for Q1 journals:** *This study is among the first to combine AI-specific (Transparency, Anxiety) and adaptivity-specific (Personalization, Feedback Quality) constructs in a single adoption model for course-like AI-adaptive platforms, applying a hybrid SEM–ANN approach to both test theory and rank factor importance — in an underrepresented developing-country context.*

> **Pre-submission check:** *Before final submission, search for recent papers combining hybrid SEM–ANN with AI anxiety in EdTech contexts (including SSRN preprints). At least one such paper has emerged in early 2025. Retrieve it, cite it, and explicitly distinguish your focus on adaptive learning platform adoption (not general AI tool adoption) from these parallel contributions.*

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6. Theoretical Foundation

Base Theory: UTAUT2 (Venkatesh et al., 2012) — the most comprehensive general technology acceptance model — serves as the theoretical anchor. UTAUT2 has been widely applied in recent AI adoption studies in higher education (Algerafi et al., 2023; Boubker, 2024; Falebita & Kok, 2024).

From UTAUT2, three core constructs are retained:

- Performance Expectancy (PE)
- Effort Expectancy (EE)
- Social Influence (SI)

UTAUT2 Constructs Intentionally Excluded

Excluded Construct	Justification
Hedonic Motivation	Academic adaptive platforms are utility-driven, not hedonic. Students use them to improve performance or skills, not for entertainment. Consistent with prior institutional e-learning adoption studies (Falebita & Kok, 2024).
Price Value	Most university-provided AI-adaptive platforms have no direct student cost. Price Value is irrelevant where there is no financial transaction.
Habit	This study focuses on *initial/early adoption intention*, not continuance or sustained use. Habit is only meaningful for behaviors repeated over time.
Facilitating Conditions	Participants are screened to confirm prior AI tool exposure, which acts as a de facto proxy for basic technological conditions being met. Recent adoption studies in academic AI contexts found FC non-significant when basic infrastructure is assumed (Boubker, 2024; Algerafi et al., 2023). Recommended for future studies in lower-infrastructure contexts.

> **Defense note on Facilitating Conditions:** *If your supervisor pushes back, cite Boubker (2024) and Algerafi et al. (2023) where FC was non-significant in academic AI adoption, and note that your screening criterion filters out participants who lack basic access. Offer to include FC as a future research direction.*

Why Trust Is Excluded

Trust is one of the most cited constructs in AI adoption models (Ngo et al., 2025; Herm et al., 2022). Its exclusion is deliberate and defensible on three grounds:

1. **Construct overlap:** AI Transparency (AT) already captures the cognitive dimension of trust — users trust systems they understand. Including a separate trust construct introduces multicollinearity risk between AT and Trust.
2. **Model parsimony:** Adding trust requires 4 additional items and two extra structural paths, making the model unnecessarily complex for a BSc thesis with a moderate sample size.
3. **DV scope:** Trust matters more for *continued use* and *actual behavior*. For *initial adoption intention*, perceived utility and ease carry greater explanatory weight (Venkatesh et al., 2012; Falebita & Kok, 2024).

Trust is explicitly listed as a limitation in Section 12 and as a priority extension for future research.

Extended with four AI/adaptivity-specific constructs (primary novel contribution):

- Perceived Personalization (PP) — grounded in Singh & Zhang (2022); Bozkurt et al. (2024); Halkiopoulos & Gkintoni (2024)
- AI Transparency (AT) — grounded in Shin & Park (2019); Shin (2021); Ngo et al. (2025)
- AI Anxiety (AA) — grounded in Wang & Wang **(2019)**; Wang et al. (2022); Li & Huang (2020) — *negative predictor*
- Feedback Quality (FQ) — grounded in Afzaal et al. (2024); Halkiopoulos & Gkintoni (2024); IJEETHE (2025)

Moderator:

- AI Familiarity (AF) — prior experience with AI tools, moderates the effect of AI Anxiety on Behavioral Intention (Wang et al., 2022; Algerafi et al., 2023)

Outcome Variable:

- Behavioral Intention to Use (BI) — standard UTAUT2 dependent variable (Venkatesh et al., 2012)

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7. Conceptual Model

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│                      CONCEPTUAL MODEL                           │
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│  UTAUT2 Core                                                  │
│  ┌─────────────────────────────┐                              │
│  │ Performance Expectancy (PE)│ ──────────────────────────────┐          │
│  └─────────────────────────────┘                              │          │
│  ┌─────────────────────────────┐                              │          │
│  │ Effort Expectancy (EE)      │ ──────────────────────────────┤          │
│  └─────────────────────────────┘                              │          │
│  ┌─────────────────────────────┐                              ▼          │
│  │ Social Influence (SI)       │ ──────────────▶  ┌──────────────────┐   │
│  └─────────────────────────────┘                  │   Behavioral     │   │
│                                                   │   Intention      │   │
│  AI/Adaptivity-Specific (Novel)                   │   to Use (BI)    │   │
│  ┌─────────────────────────────┐              ▲   └──────────────────┘   │
│  │ Perceived Personalization   │ ─────────────┤                          │
│  │ (PP)                        │              │                          │
│  └─────────────────────────────┘              │                          │
│  ┌─────────────────────────────┐              │                          │
│  │ AI Transparency (AT)        │ ─────────────┤                          │
│  └─────────────────────────────┘              │                          │
│  ┌─────────────────────────────┐              │                          │
│  │ AI Anxiety (AA) [NEGATIVE]  │ ─────────────┤  (negative path, β < 0)  │
│  └─────────────────────────────┘              │                          │
│  ┌─────────────────────────────┐              │                          │
│  │ Feedback Quality (FQ)       │ ─────────────┘                          │
│  └─────────────────────────────┘                                         │
│                                                                          │
│  Moderator: AI Familiarity (AF) moderates AA → BI path                   │
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Total constructs: 7 predictors (6 positive + 1 negative) + 1 moderator → Behavioral Intention to Use (BI)

8. Constructs & Definitions

> *All survey items use a* **5-point Likert scale** *(1 = Strongly Disagree, 5 = Strongly Agree). A 5-point scale was selected to minimize respondent fatigue given the survey length (36 items) and to align with the established UTAUT2 measurement tradition (Venkatesh et al., 2012; Algerafi et al., 2023). Novel constructs (PP, FQ) use items developed de novo informed by the literature, following Churchill's (1979) scale development procedure. This is stated explicitly in the Methods section.*

C1: Performance Expectancy (PE)

Definition: The degree to which a student believes that using the AI-adaptive learning platform will improve their learning effectiveness — whether in formal academic coursework or in self-directed learning outside a classroom setting.

Theoretical base: Venkatesh et al. (2012) UTAUT2 — *primary scale*; applied in AI EdTech contexts by Falebita & Kok (2024); Al-Adwan et al. (2023)

Note on scope: PE is intentionally defined to cover both academic and self-directed learners. AI-adaptive platforms are used in both contexts (e.g., Duolingo for self-study, ALEKS for coursework), and restricting PE to academic outcomes alone would exclude a meaningful segment of adoption motivation.

Survey items (3 items):

- PE2: "I believe AI-adaptive platforms would improve my learning effectiveness, both in my coursework and in self-directed learning."
- PE3: "Using an AI-adaptive platform would help me achieve my learning goals faster."
- PE4: "I find AI-adaptive learning tools useful for acquiring knowledge in areas I find difficult."

> **Why PE1 was dropped:** *PE1 ("helps me learn course material more effectively") was too similar to PE2 and narrowly academic in framing. PE2 revised now serves as the broader anchor item covering all learning contexts.*

C2: Effort Expectancy (EE)

Definition: The degree of ease associated with using the AI-adaptive learning platform.

Theoretical base: Venkatesh et al. (2012) UTAUT2 — *primary scale*; applied in AI contexts by Falebita & Kok (2024); Salar & Hamutoglu (2022)

Survey items (3 items):

- EE1: "My interaction with an AI-adaptive learning platform is clear and easy to understand."
- EE2: "It is easy for me to navigate through an AI-adaptive learning platform."
- EE3: "Learning how to use an AI-adaptive platform requires little effort from me."

> **Why EE4 was dropped:** *EE4 ("I would find AI-adaptive platforms easy to use overall") was a generic summary sentence that added no new facet beyond what EE1, EE2, and EE3 already captured. EE1 covers clarity of interaction, EE2 covers navigation, EE3 covers learning effort — three distinct facets.*

C3: Social Influence (SI)

Definition: The degree to which a student perceives that important others — peers, instructors, or family — believe they should use the AI-adaptive platform.

Theoretical base: Venkatesh et al. (2012) UTAUT2 — *primary scale*; Algerafi et al. (2023); Boubker (2024)

Bangladesh relevance: Social Influence is expected to be a particularly strong predictor in Bangladesh given the collectivist educational culture, where peer recommendations and instructor endorsement carry significant normative weight over individual adoption decisions.

Survey items (3 items):

- SI1: "People important to me (family, close friends) think I should use AI-adaptive learning platforms."
- SI2: "My instructors encourage or recommend using AI-adaptive tools for studying."
- SI3: "Many of my classmates use and positively recommend AI-adaptive learning platforms."

C4: Perceived Personalization (PP) ⭐ Novel

Definition: The degree to which a student perceives that the AI-adaptive platform genuinely tailors its content sequencing, difficulty, and learning path to their individual needs — rather than delivering a fixed or generic curriculum.

Theoretical base: Singh & Zhang (2022) — *Computers & Education: AI*; Bozkurt et al. (2024); Halkiopoulos & Gkintoni (2024); Gligorea et al. (2023)

Scale development note: No validated adoption-context scale for Perceived Personalization currently exists. Items were developed de novo following Churchill's (1979) scale development procedure, informed by the adaptive learning and personalization literature. This is explicitly stated in the Methods section and positions PP as a scale contribution of this study.

Construct boundary: PP measures perceived personalization of *content, difficulty, and learning path*. It does not measure feedback quality, which is captured separately by FQ. This distinction is critical for discriminant validity between PP and FQ.

Survey items (4 items):

- PP1: "The AI-adaptive platform tailors its learning content specifically to my individual level and pace."
- PP2: "The platform adjusts my learning path based on my previous performance, not a fixed curriculum." *(revised — replaces original PP2 which overlapped with FQ3)*

- PP3: "I feel that the platform truly understands and responds to my personal learning needs."
- PP4: "The difficulty of tasks and exercises adjusts according to my actual performance in real time."

> **Why PP2 was revised:** *The original PP2 ("feedback relevant to my specific mistakes, not generic advice") overlapped almost identically with FQ3 ("feedback specific to my individual mistakes"). This would have caused discriminant validity failure between PP and FQ (HTMT > 0.85). The new PP2 focuses on learning path adjustment — a purely personalization-side dimension with no overlap with FQ.*

C5: AI Transparency (AT) ⭐ Novel

Definition: The degree to which a student understands and can predict the reasoning behind the AI system's decisions and recommendations about their learning path.

Theoretical base: Shin & Park (2019) — *Computers in Human Behavior*, 98, 277–284 — *primary foundational source*; Shin (2021) — *International Journal of Human-Computer Studies*, 146, Article 102551; Ngo et al. (2025); Herm et al. (2022)

Survey items (4 items):

- AT1: "I understand why this AI platform recommends specific content or topics to me."
- AT2: "The AI platform clearly explains how it personalizes my learning experience."
- AT3: "I have a clear sense of what information the AI uses to make decisions about my learning path."
- AT4: "I can roughly predict what the platform will suggest next based on my previous performance."

C6: AI Anxiety (AA) ⭐ Novel — Negative Predictor

Definition: The degree of discomfort, apprehension, or unease a student experiences when interacting with an AI system that continuously monitors, profiles, and makes autonomous decisions about their learning process. **In this study, AI Anxiety is operationalized specifically as surveillance and profiling anxiety in adaptive learning contexts — the discomfort arising from the system's continuous evaluation and autonomous control over one's learning path — rather than general AI anxiety about job displacement or broad technological change.**

Theoretical grounding: Informed by Wang & Wang **(2019)** — *Interactive Learning Environments*, 30(4), 619–634 — who established AI anxiety as a multidimensional construct with learning, job replacement, sociotechnical blindness, and AI configuration dimensions. The present study draws most directly on the **learning** and **AI configuration** dimensions of that framework, but adapts and narrows the construct to the unique surveillance context of AI-adaptive platforms, where the AI continuously builds a profile of the student and makes decisions about their learning trajectory. Additional grounding: Wang et al. (2022); Li & Huang (2020).

> **Important methodological note on item development:** *The four AA items were developed de novo for the adaptive learning surveillance context, informed by Wang & Wang (2019) and AI-privacy literature, and are* **not** *claimed to be direct adaptations of specific AIAS subscale items (whose content covers job replacement anxiety and sociotechnical blindness — dimensions irrelevant to this context). This distinction should be stated clearly in the Methods section. During the pilot study, Cronbach's alpha for AA will be verified (target $\alpha > 0.70$); any item with item-total correlation below 0.40 will be replaced.*

> **Critical methodological note on coding:** *AA items are coded* **directly** *— a score of 5 means high anxiety. The negative relationship between AA and BI is captured entirely by the structural path coefficient ($\beta < 0$) in SmartPLS.* **Do not reverse-score these items.** *Reverse-scoring would flip the construct's meaning and produce a measurement error. This must be stated clearly in the Methods section.*

Survey items (4 items):

- AA1: "I feel uncomfortable when I know an AI system is continuously evaluating my learning performance."
- AA2: "The idea of an AI system making decisions about my learning path makes me uneasy."
- AA3: "I feel uneasy knowing the AI system builds a detailed profile of my academic strengths and weaknesses." *(sharpens profiling/surveillance discomfort angle, distinct from AA4)*
- AA4: "I feel anxious when I think about my learning data being tracked and analyzed automatically."

> **Construct boundary:** *AA3 focuses on discomfort about the AI building a model of the student; AA4 focuses on discomfort about continuous data tracking. These are related but meaningfully distinct privacy-adjacent concerns within the adaptive learning surveillance experience. The AA construct is conceptually adjacent to information privacy concern (Malhotra et al., 2004) but is grounded specifically in the AI-adaptive learning interaction — not general online privacy.*

C7: Feedback Quality (FQ) ⭐ Novel

Definition: The degree to which a student perceives the AI-generated feedback as accurate, timely, specific, and actionable for improving their understanding.

Theoretical base: Afzaal et al. (2024) — *Education and Information Technologies*; Halkiopoulos & Gkintoni (2024) — *Electronics*; IJEETHE systematic review (2025); Zheng et al. (2024)

Scale development note: No validated adoption-context scale for Feedback Quality in AI-adaptive systems currently exists. Items were developed de novo following Churchill's (1979) procedure, informed by the adaptive learning and formative feedback literature. This is stated explicitly in the Methods section.

Four facets covered: accuracy (FQ1), timeliness (FQ2), specificity (FQ3), actionability (FQ4). Each item maps to a distinct dimension of feedback quality as defined in the formative feedback literature.

Survey items (4 items):

- FQ1: "The feedback I receive from this AI platform is accurate and genuinely helpful."
- FQ2: "The AI provides feedback immediately after I make a mistake, while the content is still fresh in my mind." *(covers timeliness, a facet in the definition that had no item in prior versions)*
- FQ3: "The feedback is specific to my individual mistakes — not just a generic comment."
- FQ4: "The AI platform's feedback helps me understand my errors and correct them effectively."

Moderator: AI Familiarity (AF)

Definition: The degree of prior experience and exposure a student has had with AI-based tools, systems, or applications before using the adaptive learning platform.

Theoretical base: Algerafi et al. (2023) — *IEEE Access*; Falebita & Kok (2024); Wang et al. (2022)

Role: Moderates the relationship between AI Anxiety (AA) and Behavioral Intention (BI). Students with greater prior AI experience are expected to exhibit lower anxiety-driven resistance to adoption, weakening the negative AA → BI relationship.

Scale development note: AF items are informed by the AI familiarity operationalizations in Algerafi et al. (2023) and Wang & Wang (2019), where familiarity with AI tools is treated as an experience-based construct. All items measure prior *experience and exposure* — not comfort level, which would overlap with AA.

Survey items (3 items):

- AF1: "I have used AI-powered tools (e.g., chatbots, recommendation systems, AI tutors) before."
- AF2: "I have significant experience interacting with AI-based applications in my daily life."
- AF3: "I am familiar with how AI tools generally analyze data and make suggestions."

Dependent Variable: Behavioral Intention to Use (BI)

Definition: The strength of a student's intention to use AI-based adaptive learning platforms in their current or future academic or self-directed learning activities.

Theoretical base: Venkatesh et al. (2012) UTAUT2 — *primary scale*; Algerafi et al. (2023); Boubker (2024)

Survey items (3 items):

- BI1: "I intend to use AI-adaptive learning platforms in my current studies."
- BI2: "I plan to regularly use AI-adaptive platforms throughout my academic studies." *(minor revision — removed "in the future" which made the intention feel distant and hypothetical)*
- BI4: "I would recommend AI-adaptive learning platforms to my classmates."

> **Why BI3 was dropped:** *BI3 ("I plan to use AI-adaptive learning platforms frequently whenever they are available to me") was nearly identical to BI2 in meaning — both expressed future frequent use intention. BI2 is the cleaner, more direct statement.*

> **Note on BI4 (word-of-mouth item):** *BI4 captures recommendation intention — a distinct behavioral dimension beyond self-use intention, consistent with UTAUT2 extensions in AI EdTech contexts (Algerafi et al., 2023).*
> **Robustness check:** *During analysis, the structural model will be re-estimated using only BI1 + BI2 (excluding BI4) to confirm that path coefficients and significance levels are stable. If BI4 introduces structural instability, it will be reported as a separate dependent variable or dropped with justification.*

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9. Hypotheses

#	Hypothesis	Direction	Rationale
H1	Performance Expectancy positively influences Behavioral Intention to Use	(+)	Core UTAUT2 finding. Students who believe the platform improves their learning — in coursework or self-directed contexts — are more likely to adopt it (Falebita & Kok, 2024; Al-Adwan et al., 2023).
H2	Effort Expectancy positively influences Behavioral Intention to Use	(+)	Lower perceived effort reduces adoption barriers. Confirmed across AI EdTech contexts (Algerafi et al., 2023; Salar & Hamutoglu, 2022).
H3	Social Influence positively influences Behavioral Intention to Use	(+)	Peer and instructor recommendations are especially important in collectivist cultures. Particularly relevant to the Bangladesh higher education context (Boubker, 2024; Algerafi et al., 2023).
H4	Perceived Personalization positively influences Behavioral Intention to Use	(+)	Students who perceive the platform as genuinely adapting its content and path to them are more motivated to adopt it. This is the core value proposition of AI-adaptive systems and the study's primary novel hypothesis (Singh & Zhang, 2022; Bozkurt et al., 2024).
H5	AI Transparency positively influences Behavioral Intention to Use	(+)	Understanding why the AI makes decisions reduces "black box" apprehension and increases adoption willingness (Shin & Park, 2019; Ngo et al., 2025; Herm et al., 2022).

#	Hypothesis	Direction	Rationale
H6	AI Anxiety negatively influences Behavioral Intention to Use	(−)	Discomfort and fear of AI surveillance and profiling reduce adoption willingness. AA is coded directly (higher score = more anxiety); the negative direction is the structural path coefficient β < 0 in SmartPLS — no item reversal is performed. Grounded in Wang & Wang **(2019)**; Wang et al. (2022).
H7	Feedback Quality positively influences Behavioral Intention to Use	(+)	High-quality, timely, specific feedback is a core value driver of adaptive platforms. Students who perceive feedback as accurate and actionable are more motivated to continue using the system (Afzaal et al., 2024; Zheng et al., 2024).
H8	AI Familiarity moderates the relationship between AI Anxiety and Behavioral Intention to Use, such that the negative effect of AA on BI is weaker for students with higher AI Familiarity	(moderating)	Students with prior AI experience have more realistic expectations, reducing anxiety-driven resistance to adoption (Wang et al., 2022; Algerafi et al., 2023).

> **Most novel hypotheses:** *H4, H5, and H7 are the primary theoretical contributions. H6 is methodologically notable for including a negative predictor — uncommon in EdTech adoption models. If ANN shows PP (H4) as the single highest-importance predictor, that is the study's headline publishable finding.*

> **Note on H8:** *If moderation analysis becomes too complex for your BSc timeline, simplify to a direct AF → BI path and flag the moderation test as future work. Both are defensible.*

10. Research Methodology

10.1 Research Design

Type: Quantitative, cross-sectional, survey-based. **Unit of analysis:** Individual undergraduate and postgraduate students in Bangladeshi higher education institutions. **Approach:** Deductive — testing a theoretically grounded conceptual model with empirical survey data.

Justification for PLS-SEM over CB-SEM: PLS-SEM is selected over covariance-based SEM (CB-SEM, e.g., AMOS or lavaan) for three reasons: (1) the model is exploratory and theory-extending — integrating novel constructs (PP, FQ, AA) into an extended UTAUT2 framework — rather than strictly confirmatory, which suits PLS-SEM's predictive orientation (Hair et al., 2022); (2) PLS-SEM produces stable path estimates with moderate samples and complex models (7 predictors + 1 moderator), whereas CB-SEM fit indices become unreliable under these conditions; and (3) PLS-SEM does not assume multivariate normality, which cannot be guaranteed in a convenience sample drawn from diverse institutions. A CB-SEM robustness check may be run if requested by a reviewer, as SmartPLS 4 now supports basic CB-SEM estimation (Ringle et al., 2024).

10.2 Population & Sample

Population: Current undergraduate and postgraduate students in Bangladeshi universities and colleges who have been exposed to or used at least one AI-assisted or AI-adaptive digital learning tool.

Sample size target: 250–350 usable responses.

- Minimum 200 for PLS-SEM to achieve stable path coefficients (Hair et al., 2022).
- Aim for 300+ for ANN stability and reliable cross-validation.
- Justify via G*Power: medium effect size ($f^2 = 0.15$), $\alpha = 0.05$, power = 0.80, 7 predictors → minimum ~103 cases; target 300 for robustness.

Sampling strategy: Non-probability convenience + snowball sampling via university department emails, student WhatsApp/Facebook groups, and campus QR code posters. Acknowledged as a limitation in the paper.

Screening criteria: Respondents must confirm prior exposure to at least one AI-assisted learning tool via the Section A screening question. A plain-language description of AI-adaptive platforms is included at the survey introduction.

10.3 Survey Instrument

Section	Content	Items
A	Screening + Demographics	5
B	Performance Expectancy (PE)	3
C	Effort Expectancy (EE)	3
D	Social Influence (SI)	3
E	Perceived Personalization (PP)	4
F	AI Transparency (AT)	4
G	AI Anxiety (AA)	4
H	Feedback Quality (FQ)	4
I	AI Familiarity (AF)	3
J	Behavioral Intention to Use (BI)	3
Total		**36 items**

Scale: All construct items measured on a **5-point Likert scale** (1 = Strongly Disagree, 5 = Strongly Agree). The

5-point format was chosen to minimize respondent fatigue across 36 items and is consistent with validated UTAUT2 instruments (Venkatesh et al., 2012; Algerafi et al., 2023). Pilot Cronbach's alpha will confirm adequacy of scale variance.

Item sourcing by construct:

- **PE, EE, SI, BI:** Adapted from Venkatesh et al. (2012) and recent AI EdTech applications.
- **AT:** Adapted from Shin & Park (2019) and Shin (2021).
- **AA:** Developed de novo for the AI-adaptive learning surveillance context, informed by Wang & Wang **(2019)** (specifically the learning and AI configuration dimensions) and AI-privacy literature. Items are not claimed as direct AIAS adaptations given the AIAS's broader scope (including job replacement anxiety); rather, they operationalize a narrower, context-specific form of AI anxiety. Pilot reliability will confirm $\alpha > 0.70$.
- **PP and FQ:** Developed de novo following Churchill's (1979) scale development procedure.
- **AF:** Informed by operationalizations in Algerafi et al. (2023) and Wang & Wang **(2019)**.

Survey platform: Google Forms (free) or Qualtrics (if university provides access).

Section A — Demographics & Screening (5 items):

Code	Question
A1	**Screening:** Have you ever used, tried, or been exposed to an AI-based adaptive learning tool or platform? ☐ Yes ☐ No *[If No → end survey]*
A2	**Gender:** ☐ Male ☐ Female ☐ Prefer not to say
A3	**Age:** ☐ 18–20 ☐ 21–23 ☐ 24–26 ☐ 27 or above
A4	**Institution type and level:** ☐ Public University (Undergraduate) ☐ Public University (Postgraduate) ☐ Private University (Undergraduate) ☐ Private University (Postgraduate) ☐ National University College
A5	**Field of study:** ☐ Science/Engineering ☐ Business ☐ Social Science/Arts ☐ Other

Survey introduction paragraph (show to all participants before Section A):

> *"This survey is about AI-based adaptive learning platforms — digital systems that automatically adjust the content, difficulty level, and feedback you receive based on your performance. Unlike regular e-learning tools such as Moodle or Google Classroom, these platforms learn from your responses and make personalized decisions about what you should study next — whether you are using them as part of a university course or learning on your own. For example, if you answer a question incorrectly, the system automatically provides an easier version of the same concept. Examples include ALEKS, Knewton Alta, and Duolingo's adaptive engine.* **If you have used one of these platforms directly, please answer based on that experience. If you have not yet used such a platform but are familiar with AI tools generally, please answer based on your expectations and perceptions of how such a system would work, using the description above as your guide.** *All responses are fully anonymous and used only for academic research."*

10.4 Data Analysis: Hybrid PLS-SEM + ANN

STAGE 1 — PLS-SEM (SMARTPLS 4)

Step 1: Measurement Model Assessment

Check	Threshold
Factor loadings	≥ 0.70
Composite Reliability (CR)	≥ 0.70
Average Variance Extracted (AVE)	≥ 0.50
HTMT ratio (discriminant validity)	< 0.85 (strict) or < 0.90 (lenient)
Fornell–Larcker criterion	AVE > squared inter-construct correlations

Step 2: Common Method Bias (CMB) Assessment

Two complementary approaches:

1. **Harman's Single Factor Test** (SPSS): All items loaded onto one factor; if it explains < 50% of variance, CMB is not severe. This is a detection check, not a control.
2. **Common Latent Factor (CLF) method** (SmartPLS 4): Add an unmeasured latent factor with paths to all indicators. Compare path coefficients with and without CLF. If differences < 0.200, CMB is not a major concern. This is the recommended statistical remedy for PLS-SEM (Hair et al., 2022).
3. **Procedural remedies** (data collection): full anonymity, voluntary participation, randomized item order within constructs, clear separation of predictor and outcome items.

Step 3: Structural Model Assessment

Output	Purpose
Path coefficients (β) + p-values	Hypothesis testing (bootstrapping: 5,000 subsamples)
R^2 of BI	Variance explained in the dependent variable
Effect size f^2	Practical significance of each path
Predictive relevance Q^2	Blindfolding; $Q^2 > 0$ = predictive relevance confirmed
PLSpredict (out-of-sample)	Compare PLS-SEM RMSE and MAE against a naïve linear model benchmark; confirms out-of-sample predictive power beyond in-sample Q^2 (Hair et al., 2022)

Step 4 (if testing H8): Moderation Analysis

- Interaction term: AI Familiarity × AI Anxiety → BI

- Created using SmartPLS product-indicator or two-stage approach
- Report β, t-value, and 95% bias-corrected confidence interval for the interaction term

STAGE 2 — ANN (PYTHON OR SPSS)

Goal: Complement PLS-SEM with nonlinear importance ranking of all predictors.

- **Inputs:** Latent variable scores exported from SmartPLS (one composite score per construct per respondent).
- **Target variable:** BI latent variable scores.
- **Architecture:** Multi-Layer Perceptron (MLP), one hidden layer, **ReLU activation** (hidden layer) + **linear activation** (output layer). ReLU is the current standard for regression-type MLPs; sigmoid activation in hidden layers causes vanishing gradients and is no longer recommended. The output layer uses linear (identity) activation since BI scores are continuous. In SPSS Neural Networks, select "Identity" for the output layer activation.
- **Validation:** 10-fold cross-validation to prevent overfitting.
- **Key outputs:** (a) Normalized importance scores (sum to 100%) — a ranked bar chart of construct importance; (b) **ANN R^2** and **RMSE** on held-out test folds, reported alongside PLS-SEM R^2 to demonstrate convergent predictive validity between the two methods.

Python: `scikit-learn` (MLPRegressor) preferred. `tensorflow/keras` as alternative. **SPSS:** Analyze → Neural Networks → Multilayer Perceptron.

How to present results:

> *"PLS-SEM identified Perceived Personalization (β = 0.XX, p < 0.001) and AI Transparency (β = 0.XX, p < 0.001) as the two strongest positive predictors. ANN confirmed PP as the single most important predictor (normalized importance = XX%), followed by AT (XX%), providing convergent evidence and suggesting PP's effect may be nonlinear — underestimated by linear path analysis alone."*

10.5 Software

Tool	Purpose
SmartPLS 4	PLS-SEM — **use the 30-day Professional trial for analysis** (the free Student License is limited to fewer than 100 observations and cannot accommodate the target sample of 250–350). Time the trial start to coincide with clean data ready for analysis.
Python (scikit-learn)	ANN modeling, cross-validation, normalized importance chart
SPSS	Descriptive statistics, Harman's test, pilot Cronbach's α
Google Forms	Survey distribution

Tool	Purpose
G*Power	Sample size justification
draw.io / PowerPoint	Conceptual model diagram
Zotero	Reference management

10.6 Ethical Considerations

- Informed consent from all participants before survey access.
- Full anonymity — no personal identifiers collected.
- Voluntary participation — withdrawal at any time without penalty.
- Ethical approval obtained from the relevant **departmental or faculty research ethics committee** prior to data collection, in accordance with the institution's research governance procedures. ("IRB" is US-specific terminology; the Bangladeshi equivalent is typically a departmental or faculty-level committee.)
- Data used solely for academic research, not shared with third parties.

11. Expected Contributions

Theoretical contributions:

- Proposes and tests a novel adoption model integrating UTAUT2 with four AI/adaptivity-specific constructs (PP, AT, AA, FQ) in the specific context of course-like AI-adaptive platforms, applicable to both formal academic and self-directed learning contexts.
- Demonstrates the value of the hybrid SEM–ANN approach in EdTech adoption research — SEM for theory testing, ANN for nonlinear importance ranking.
- Contributes validated scale items for novel constructs (PP, FQ, and AF in the AI-adaptive context) usable by future researchers.
- Operationalizes AI Anxiety narrowly as surveillance and profiling anxiety specific to adaptive learning environments — a conceptual contribution that distinguishes this form of anxiety from general AI anxiety scales.
- Provides evidence from an underrepresented developing-country context (Bangladesh).

Practical contributions:

- Delivers a ranked list of factors influencing adoption. If ANN shows PP as the strongest predictor, platform designers should prioritize personalization quality above interface simplicity or social promotion.
- Guidance for Bangladeshi universities and EdTech developers on which barriers to address.

Policy contribution:

- Highlights AI-specific barriers (anxiety, transparency deficits) that policymakers should address when mandating or promoting adaptive learning in higher education.

12. Limitations & Future Directions

Limitations:

- **Self-report bias:** Perceptions and intentions may not reflect actual behavior. Future studies should incorporate behavioral usage logs.
- **Cross-sectional design:** Cannot establish causality. A longitudinal design would capture intention → actual use dynamics.
- **Convenience sampling:** Limits generalizability beyond the sample.
- **Single-country context:** Findings may not generalize to other cultures or education systems.
- **Exposure vs. direct use:** Some participants may have limited direct experience with AI-adaptive platforms specifically, and will have responded based on perceptions or expectations informed by general AI tool experience. This is standard and acceptable in adoption intention research (TAM/UTAUT studies routinely measure intention before use), but may reduce measurement precision for novel construct items (PP, FQ, AA) that presuppose familiarity with adaptive-specific features.
- **Trust excluded:** Trust is a recognized predictor of AI adoption (Ngo et al., 2025) and was excluded for parsimony and to avoid multicollinearity with AT.
- **Facilitating Conditions excluded:** FC may matter in lower-infrastructure sub-regions of Bangladesh.
- **AA construct scope:** AI Anxiety is operationalized narrowly as surveillance/profiling anxiety. Broader dimensions of AI anxiety (job replacement, sociotechnical blindness per Wang & Wang, 2019) are outside the scope of this study and should be explored in future research.

Future directions:

- Longitudinal study to capture actual adoption behavior and continuance over time.
- Multi-country comparison (Bangladesh vs. India vs. Malaysia) to test cultural boundary conditions.
- Mixed-methods approach adding qualitative interviews to explain unexpected SEM findings.
- Testing the model with actual behavioral data (platform usage logs) rather than self-reported intention.
- Including Trust as a mediator between AI Transparency and BI.
- Replicating with Facilitating Conditions added for lower-infrastructure populations.
- Expanding AI Anxiety to include broader AIAS dimensions (job replacement, sociotechnical blindness) in contexts beyond adaptive learning platforms.
- Extension to secondary education students and professional learners.

13. Action Plan & Timeline

Phase	Activities	Target
Phase 1 — Preparation	Finalize constructs and hypotheses; supervisor approval; ethics clearance	Weeks 1–2
Phase 2 — Instrument Design	Draft full 36-item questionnaire; supervisor review	Weeks 3–4
Phase 3 — Pilot Study	Distribute to 30–50 students; Cronbach's α; remove items with loading < 0.70	Weeks 5–6
Phase 4 — Full Data Collection	Distribute via emails, WhatsApp groups, QR posters; target 300 responses; 2 reminders	Weeks 7–10
Phase 5 — Data Cleaning	Remove invalid responses; check for outliers; **non-response bias check** — compare early vs. late respondents on key construct means (Armstrong & Overton, 1977); if no significant differences, bias is not a concern	Week 11
Phase 6 — PLS-SEM Analysis	Activate SmartPLS 30-day Professional trial; measurement model → CMB → structural model → bootstrapping	Weeks 12–13
Phase 7 — ANN Analysis	Export latent scores; train MLP; 10-fold CV; normalized importance chart	Week 14
Phase 8 — Writing	Methods → Results → Discussion → Introduction → Abstract → References	Weeks 15–19
Phase 9 — Defense Prep	Slides, mock Q&A, supervisor final review	Week 20
Phase 10 — Journal Submission	Format for target journal; cover letter; literature check for competing papers; submit	Week 21+

> **Defense tip:** *Anchor every result to (1) your operational definition — course-like autonomous adaptation, not a chatbot — and (2) your hybrid method rationale — SEM tests theory, ANN reveals nonlinear importance. Consistency between your rationale and findings is what committee members look for.*

14. Tools & Target Journals

Target Journals (Ranked by Fit)

Rank	Journal	Publisher	Notes
1	*Computers & Education: Artificial Intelligence*	Elsevier	Perfect scope; publishes SEM-ANN EdTech papers; faster review cycle. **Open-access; APC USD 2,940 (current 2025 figure) — apply for Elsevier's developing-country APC waiver program**
2	*Education and Information Technologies*	Springer	Strong fit; publishes hybrid-method adoption studies
3	*Computers in Human Behavior*	Elsevier	Broader audience; high impact; competitive
4	*Internet and Higher Education*	Elsevier	Excellent fit; well-regarded in EdTech adoption
5	*Telematics and Informatics*	Elsevier	Publishes IS/EdTech adoption studies
Backup	*SAGE Open*	SAGE	Open access; lower barrier for first publication

> **APC note:** *The APC for Computers & Education: Artificial Intelligence is currently* **USD 2,940** *(excluding taxes) as of 2025 per the Elsevier journal page. This is significantly higher than older figures that circulated online (~$1,800). Contact the editorial office directly about Elsevier's Research4Life waiver program, which covers authors from eligible low- and middle-income countries including Bangladesh.*

Recommended Search Keywords

Core topic: `"AI adaptive learning" adoption SEM` · `"adaptive learning platform" student acceptance UTAUT` · `PLS-SEM ANN EdTech hybrid`

Novel constructs: `"perceived personalization" e-learning adoption` · `"AI transparency" trust education technology` · `"AI anxiety" technology acceptance higher education` · `"feedback quality" adaptive learning`

Bangladesh context: `e-learning adoption Bangladesh university` · `technology acceptance developing country higher education`

Key Databases: Google Scholar → Scopus → ScienceDirect → SpringerLink → Web of Science → IEEE Xplore → SSRN (for preprints)

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15. Change Log

v1 → v2 Changes

#	Section	Change	Reason
1	C6 (AA)	Removed "reverse-scored" label from all AA items	AA items are coded directly. The negative relationship is the structural β, not item reversal.
2	AF	Removed item: *"I am comfortable working with AI-based technology in general"*	Semantically equivalent to a reversed AA item; risked HTMT > 0.85 between AF and AA.
3	BI	Removed original BI3 (comparative preference item)	Measured preference between two platforms, not pure behavioral intention.
4	Section 6	Strengthened Facilitating Conditions exclusion justification	Original justification too thin. Added Boubker (2024) and Algerafi et al. (2023).
5	Section 10.4	Added CLF method to CMB assessment	Harman's alone insufficient for Q1 reviewers (Hair et al., 2022).
6	Section 6	Added explicit Trust exclusion justification	Three-reason argument: construct overlap with AT, parsimony, DV scope.
7	All constructs	Updated all references to 2022–2025 literature	Original references too old for Q1 primary justification.
8	Section 1	Added HolonIQ (2023) statistic and UGC placeholder	Original flagged missing statistics as a "next step."
9	H6	Added explicit note that AA is direct-coded negative predictor	Prevents ambiguity during SmartPLS data entry.
10	Survey intro	Expanded with plain-language adaptive behavior example	Improves participant comprehension of what platforms are being studied.

v2 → v3 Changes

#	Section	Change	Reason
11	A4 + A5	Merged postgraduate into A4 (institution type); dropped A5 (year of study)	Year of study is partially redundant with institution type. Postgraduate status is captured more cleanly within A4. Net: −1 item.
12	C1 (PE)	Dropped PE1; revised PE2 to cover academic AND self-directed learning	PE1 narrowly framed around formal "course material." PE2 revised broadens scope to all learners — consistent with the operational definition which includes self-directed use (e.g., Duolingo). Net: −1 item.
13	C2 (EE)	Dropped EE4	EE4 was a generic summary sentence adding no new facet. EE1/EE2/EE3 cover clarity, navigation, and learning effort distinctly. Net: −1 item.

#	Section	Change	Reason
14	C4 (PP)	Replaced PP2 with path-adjustment item	Original PP2 ("feedback relevant to my specific mistakes") was semantically identical to FQ3, causing discriminant validity risk. New PP2 focuses purely on learning path adjustment. No item count change.
15	C6 (AA)	Rewrote AA3 to sharpen profiling angle	Original AA3 and AA4 were too similar. Revised AA3 focuses on AI building a profile of you; AA4 on continuous tracking.
16	C7 (FQ)	Rewrote FQ2 to cover timeliness	Original FQ2 overlapped with FQ4. Revised FQ2 covers timeliness — a facet in the FQ definition but previously without an item.
17	BI	Dropped BI3; minor revision to BI2	BI3 and BI2 were near-identical. BI2 revised removes "in the future." Net: −1 item.
18	Section 8	Added Churchill (1979) scale development framing for PP, FQ, AF	PP and FQ are novel scales with no prior validated adoption-context counterparts.
19	Sections 3 & 9	Added self-directed learner scope throughout	PE was too academically narrow. Broadened to include self-directed learning contexts consistently.

v3 → v4 Changes (This Version)

#	Section	Change	Reason
20	All	Changed "Wang & Wang (2022)" → **"Wang & Wang (2019)"** throughout	The paper was published online 14 October 2019 (received Feb 2019, accepted Aug 2019). The journal volume 30(4) is dated 2022 but the standard citation year is 2019, which is used by the overwhelming majority of citing papers. Using 2022 signals to reviewers the source was not consulted directly.
21	C6 (AA) + Section 10.3	Reframed AA item sourcing — items are **developed de novo** informed by Wang & Wang (2019), not claimed as AIAS adaptations	The full AIAS has 21 items across four dimensions (learning, job replacement, sociotechnical blindness, AI configuration). The plan's four AA items focus on evaluation and profiling anxiety — they map loosely to "learning" and "AI configuration" but not to job replacement or sociotechnical blindness. Claiming these are "adapted from AIAS" would be detected as a mismatch by domain-expert reviewers.
22	C6 (AA)	Added scope sentence to AA definition narrowing construct to surveillance/profiling anxiety	Prevents reviewers from questioning why the broader AIAS dimensions are absent; anticipates and pre-empts the critique.

#	Section	Change	Reason
23	C6 (AA)	Added note on pilot reliability check for AA items ($\alpha > 0.70$, item-total > 0.40)	Because AA items are de novo, empirical pilot validation is especially important.
24	Section 10.5 (Tools)	Updated SmartPLS note — **30-day Professional trial required for analysis**	The free Student License is limited to fewer than 100 observations. Target sample of 250–350 exceeds this limit. The trial must be activated when clean data is ready, not earlier.
25	Section 14 (Journals)	Updated APC from ~$1,800 to **USD 2,940**	Current figure from Elsevier's official journal page for *Computers & Education: Artificial Intelligence* (2025). The $1,800 figure is outdated.
26	Section 8 (all constructs)	Added one-line 5-point Likert scale justification	Q1 reviewers may query scale choice; pre-emptive justification reduces revision rounds.
27	BI (Section 8)	Added robustness check note for BI4 (re-run model with BI1+BI2 only)	BI4 is a WOM item distinct from self-use intention; confirming structural stability without it is methodologically prudent.
28	Section 5 & Section 14	Added pre-submission literature check note for competing SEM–ANN + AI anxiety papers	A recent SSRN preprint (early 2025) applies a similar method to general AI tool adoption. Must be cited and distinguished before final submission.
29	C6 (AA)	Added note on AA's conceptual adjacency to information privacy concern literature	AA3 and AA4 are privacy-adjacent; acknowledging this and citing Malhotra et al. (2004) or equivalent strengthens grounding.
30	Section 12 (Limitations)	Added AA construct scope as an explicit limitation	Honest acknowledgment that broader AI anxiety dimensions are out of scope strengthens the paper's credibility.
31	AF (Section 8)	Updated AF reference from "Wang & Wang (2022)" to "Wang & Wang (2019)"	Consistent with change #20.

v4 → v5 Changes (This Version)

#	Section	Change	Reason
32	Section 10.1	Added PLS-SEM vs. CB-SEM justification paragraph	Q1 reviewers expect explicit rationale for method choice; also flags CB-SEM robustness check as optional.

#	Section	Change	Reason
33	Section 10.4 (ANN)	Corrected activation function: sigmoid → **ReLU** (hidden), **linear** (output)	Sigmoid in hidden layers causes vanishing gradients; ReLU is the current standard for MLP regression tasks. Incorrect activation is an immediate technical red flag for quantitative reviewers.
34	Section 10.4 (ANN)	Added ANN R^2 and RMSE to reporting plan	Normalized importance alone is insufficient; predictive accuracy metrics demonstrate the ANN adds value beyond SEM and allow comparison of R^2 across both methods.
35	Section 10.4 (PLS-SEM)	Added PLSpredict step to structural model assessment	Out-of-sample predictive assessment is now a Q1 standard expectation per Hair et al. (2022); Q^2 alone is in-sample only.
36	Section 10.5 (Tools)	Added Ringle et al. (2024) software citation with full reference	SmartPLS requires its own citation separate from Hair et al. (2022); journals that publish SEM studies routinely check for this. Also noted Hair et al. (2022) is for SmartPLS 3, and the 4th edition is forthcoming in 2026.
37	Section 10.3 (Survey intro)	Updated to address perception vs. experience gap	Participants with no direct adaptive platform experience are now instructed to respond based on expectations; this pre-empts reviewer concern about response validity and aligns with standard TAM/UTAUT methodology.
38	Section 12 (Limitations)	Replaced vague "exposure vs. deep use" bullet with specific perception-based response limitation	More precisely describes the actual measurement risk and contextualizes it within standard adoption intention research practice.
39	Section 10.6 (Ethics)	Replaced "IRB" with "departmental or faculty research ethics committee"	IRB is US-specific. Using it in a Bangladesh context is terminologically incorrect and may confuse a local committee or examiner.
40	Section 13 (Timeline)	Added non-response bias check (Armstrong & Overton, 1977) to Phase 5	Standard survey-based adoption study practice; early vs. late respondent comparison is a quick, expected quality check.
41	Key References	Added Ringle et al. (2024) and Armstrong & Overton (1977)	Both are now cited in the body of the document.
42	Key References	Added verification flag to Singh & Zhang (2022)	PP's primary theoretical grounding; must be confirmed before submission.

Quick Reference — What Makes This Study Strong

NOVELTY

✓ First adoption study focused specifically on course-like AI-adaptive platforms (not chatbots, not generic AI tools) — applies to formal and self-directed learners

✓ Integrates AI-specific (Transparency, Anxiety) + adaptivity-specific (Personalization, Feedback Quality) constructs in one model

✓ SEM–ANN hybrid in this exact context — no prior peer-reviewed paper has done this

✓ Includes negative predictor (AI Anxiety) — more realistic than positive-only models

✓ Operationalizes AI Anxiety narrowly as surveillance/profiling anxiety — a context-specific conceptual contribution beyond the general AIAS framework

✓ Contributes novel validated scales (PP, FQ, AF in AI-adaptive context)

INSTRUMENT

✓ 36 items — lean, respondent-friendly, low fatigue risk

✓ 5-point Likert scale — consistent with UTAUT2 tradition; justified for survey length

✓ All four-item novel constructs retain safety buffer for pilot loadings

✓ Clear conceptual separation between PP (path/content) and FQ (feedback quality)

✓ AA items correctly coded as direct negative predictor — no reversal errors

✓ AA sourcing correctly framed as de novo (not misrepresented as AIAS adaptations)

METHOD

✓ PLS-SEM (SmartPLS 4, 30-day Professional trial for analysis)

✓ CMB controlled via Harman's + CLF (two-method approach — Q1 standard)

✓ ANN importance ranking — nonlinear complement to SEM path coefficients

✓ AA $\beta < 0$ in structural model — methodologically correct

✓ BI4 robustness check planned

CITATIONS

✓ Wang & Wang (2019) — correct publication year used throughout

✓ Shin & Park (2019) — Computers in Human Behavior, 98, 277–284 — confirmed

✓ Shin (2021) — IJHCS, 146, 102551 — confirmed

✓ All references updated to 2022–2025 for Q1 currency

CONTEXT

✓ Bangladesh — underrepresented in high-quality AI-adoption EdTech literature

✓ Post-ChatGPT era — most relevant moment for AI-specific adoption constructs

✓ University students + self-directed learners — broader adoption relevance

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OUTPUT
✓ Ranked importance of constructs (actionable for developers and policymakers)
✓ Novel scale items for PP, FQ, AF (usable by future researchers)
✓ Actionable recommendations for platform design and institutional policy
```

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This is a living working draft (v4). Update as you read more papers, receive pilot data, and incorporate supervisor feedback. Do not share publicly — it contains your research strategy.

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